Quick Sort by selecting pivot element

(i) as first element

S.1 :

157	110	147	122	111	149	151	141	123	112	117	133

p

| 110 | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 133 | 157 |

S2:

| 110 | | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 133 | 157

p

S3:

110 | 122 | 111 | 141 | 123 | 112 | 117 | 133 | 147 | 149 | 151 | 157

p p

S4:

110 | 111 | 112 | 117 | 122 | 141 | 123 | 133 | 147 149 151 157.

p p

S : 5

110 111 | 112 | 117 | 122 | 123 | 133 | 141 147 149 151 157

p p

S : 6

| 110 | 111 | 112 | 117 | 122 | 123 | 133 | 141 | 147 | 149 | 151 | 157 |

ii) by last element

| 157 | 110 | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 133 |

S1:

| 110 | 122 | 111 | 123 | 112 | 117 | 133 | 157 | 147 | 149 | 151 | 141 |

p p

S2:

| 110 | 111 | 112 | 117 | 122 | 133 | 133 | 141 | 157 | 147 | 149 | 151 |

p p p

S3: 110 | 111 | 112 117 122 123 133 141 | 147 149 | 151 | 157

S4: 110 | 1.11 | 122 | 117 | 122 | 123 | 133 | 141 | 147 | 149 | 151 | 157

q(ii) by random element

157 | 110 | 147 | 122 | 111 | 149 | 151 | 141 | 123 | 112 | 117 | 133

S1: 110 | 122 | 111 | 112 | 117 | 123 | 157 | 147 | 149 | 151 | 141 | 133

p p

S2: 110 | 111 | 122 | 112 | 117 123 | 147 | 149 | 141 | 133 | 151 | 157

p p p p

S3: 110 111 112 | 122 | 117 | 123 | 147 | 149 | 133 | 149 151 157

p p

S4: 110 111 112 117 122 123 | 141 | 133 | 147 149 151 157

p

Quick select

Given :- [157, 110, 197, 122, 111, 199, 151, 141, 123, 112, 117, 133]

S1 : Choose pivot element

pivot = 133 [last element]

We should rearrange the array based on the logic that

→ Elements < 133 to the left of 133 and elements > 133 to the right of 133

~~Step~~

S2 : Check the index.

After partitioning

[110, 122, 111, 123, 112, 117, 133, 141, 157, 151, 199]

The index of pivot = 6

pivot index > k [6 > 2]

[110, 122, 111, 123, 112, 117]

next pivot = 117

After partition

[110, 111, 112, 117, 122, 123]

pivot index = 3 ; pivot index > k index [3 > 2]

Consider left sub array
[110, 111, 112]

next pivot = 112

after partitioning,
[110, 111, 112]

new pivot = 112

after partitioning
[110, 111, 112]

Pivot index = 2
pivot index = k index [2 = 2]

third smallest number = 112